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CO5151 – Advanced Agentic Artificial Intelligence

# D2 Interim Report

## LuatCheck: A Local, Source-Grounded Assistant over Vietnamese Regulations

Application Track – Group 8

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|---|---|
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**Repository:** https://github.com/tuandung222/LuatCheck-D2

**Reported revision:** `0928bc5fc43722187e74e94e0861ad301cba3ab1`

**Access:** public repository (no login needed)

HO CHI MINH CITY, OCTOBER 2026

**Executive Summary.** LuatCheck is a local, single-agent regulatory assistant over HCMUT academic regulations and Vietnamese labour law, driven by a 4B controller (2B option) via `llama.cpp`. It combines hybrid retrieval, cross-encoder reranking, pre-model regex guardrails, and a deterministic Python citation verifier ($< 2$ ms, zero LLM calls) with bounded repair. The D2 evaluation establishes a reproducible baseline across 1,090 trials per controller on two 50-question benchmarks (validated by a 60-item blind human audit, $\kappa = 0.80$–$0.93$), ahead of broader unseen and adversarial experiments in D3. Scaling from 2B to 4B raises strict success from 28.0–28.7% to 56.7–60.7% (+28.7 to +32.0 pp); the L1 citation verifier cuts unverified citations from 25.6–28.0% to 0.0% and lifts audited strict success from 55.3% to 60.7–64.0% (HCMUT) and 51.3% to 56.7% (labour law).

**Report Outline.**

1. Changes from the Proposal p. 1
2. Implemented Architecture p. 1
3. Tools and Permissions p. 4
4. Memory, Corpora & Retrieval Probe p. 4
5. Evaluation Set and First Results p. 5
6. Threat Model p. 8
7. Weekly Plan to D3 Presentation p. 8
8. Contributions & AI-Use Statement p. 10

## 1 Changes from the Proposal

**Scope and users.** The proposal (Application Track) targeted employees and HR staff asking labour-law questions. For this report we added a second corpus, the academic-affairs regulations of HCMUT used by students and advisers, to evaluate whether the same codebase serves an additional domain without code modifications. Prompts, tool names, and document-ID patterns are configured per domain.

Our current evaluation covers ten HCMUT academic documents and ten Vietnamese labour-law documents (the 2019 Labour Code, two related laws, six decrees, and one circular from 2020–2025), indexed from static public snapshots without tracking subsequent amendments. LuatCheck is a research prototype, does not provide legal advice, and has not yet been evaluated with end users.

Intended scenarios: a student looks up an academic rule and inspects the cited provision; an HR officer verifies a labour-law answer against the cited article; or a user pastes a document containing prompt-injection instructions, where the system must enforce its read-only tool boundary and either refuse or request clarification.

**Model, retrieval and evaluation adjustments.** Instead of the Qwen3 4B/8B controllers and 32B comparator in the proposal, LuatCheck uses two ternary-quantised local models (Bonsai 4B as default and Bonsai 2B as a lighter option) served by `llama.cpp` so that both the live demo and the evaluated controller run on a single 16 GiB laptop. For retrieval, first-stage BM25 and word 1–3-gram TF-IDF fused by reciprocal rank fusion (RRF) feed a Vietnamese cross-encoder reranker (`top_k=5`), with query anchoring on the user's original question on the first search; a dense vector index is implemented and scheduled for ablation in Week 46. Because the proposal's external held-out pair mapping could not be established, we constructed two hash-locked 50-question benchmarks and verified our offline LLM judge with a 60-item blind human audit.

**Commitments and their state.** Table 1 compares the proposal's commitments due at this checkpoint with their current state. Because no revision requests were raised at the proposal pitch, all adjustments above stem from our implementation and empirical evaluation findings.

## 2 Implemented Architecture

LuatCheck is a single-agent retrieval-augmented generation [6] system built around a bounded ReAct loop [8] (Figure 1). In Phase 1 (offline), official PDFs are extracted, normalised to Unicode NFC, segmented at article/clause boundaries with SHA-256 and page provenance, and indexed into an in-memory BM25 and word 1–3-gram TF-IDF index. In Phase 2 (online), pre-model input guardrails (shared and per-domain regular-expression patterns) screen out unsafe requests and wrap the user input as untrusted text. At each step the controller selects one JSON-constrained action: `search`, `read`, `answer`, `clarify`, or `refuse`. A search ranks sections with BM25 and TF-IDF fused by reciprocal rank fusion (RRF), rescores the top 20 candidates with a neural cross-encoder reranker, and returns the top five clauses; a read action fetches the full section text from the normalized corpus.

**Rule-based citation verifier (L1 lexical heuristic, not a model or agent).** Every proposed answer goes through a deterministic Python heuristic module ($< 2$ ms on CPU, zero LLM or neural calls) that enforces three hard gates and one diagnostic check: (i) *retrieval provenance*: every cited `article_id`

| Proposal commitment | State | Change and reason |
|---|---|---|
| Budgeted loop, schema output, citation check | Done | Runs end to end; extended with bounded repair and better quote matching. |
| Qwen3 4B/8B controllers, 32B comparator | Dropped | Replaced by two small ternary models (4B and 2B, Section 2): one 16 GiB laptop runs the demo and the benchmark model. |
| BM25 + Qdrant dense retrieval; 200 held-out question pairs | Changed | BM25 + TF-IDF with rank fusion and a neural reranker. Dense index implemented, not evaluated. External pair mapping unavailable; replaced by two 50-question benchmarks. |
| $\geq$10 end-to-end tasks, 3 configurations $\times$ 3 seeds | Done | 100 questions, 3 configurations $\times$ 3 seeds, 1,090 trials per system. |
| Reranker (optional) | Done | In the current system; ablated in Section 5. |
| Semantic (NLI) check, optional | In progress | Implemented, not used in any measurement yet. |
| Guarded summary sending, UI, audit log | Done | Changed: the model can only propose; a trusted UI approves; web UI with cited sources and agent timeline. |
| Verifier audit, 30 pairs | Done | Changed to a blind human audit of the LLM judge (30 items per benchmark, 60 total): 96.7% agreement ($\kappa = 0.93$) on HCMUT and 90.0% ($\kappa = 0.80$) on labour law. |
| Threat model | Done | Updated in Section 6. |
| First injection results | Not started | Not required at this checkpoint; planned for week 47. |
| HR walkthrough | Not started | Scheduled for week 45 (2–8 November; owners: Quyền, Phong). |

**Table 1:** Proposal commitments due at this checkpoint and their state.

must exist in the corpus and have been returned by `search` or `read_document` during the current trial; (ii) *normalised verbatim/elided quote match*: after Unicode NFC, case and whitespace/line-wrap folding, every non-trivial segment ($\geq$ 12 characters or $\geq$ 3 words) of an elided quote (`...`) must appear verbatim and in order inside the cited record, with digit-boundary guards so `5` cannot match inside `15` or `3,5`; (iii) *claim-local number and unit match*: every numeric quantity and unit in an answer sentence must occur inside the $\leq$ 300-character source window that best matches the surrounding $\pm$8 context words of that claim; and (iv) *content-word overlap*: if fewer than 50% of a sentence's non-stopword tokens appear in the best source window, a soft diagnostic warning is logged. If any of the three hard gates fails, a structured tool report naming the failed rule and quoting the closest 170-character source span is returned to the controller for at most 2 repairs; if repairs are exhausted, the loop terminates as a refusal. This provides deterministic external feedback rather than LLM self-critique [5]. The loop is bounded to 8 steps, 5 tool calls, 2 repairs and 180 s. A backend or format failure terminates the run as an execution error rather than an answer or refusal.

Using a single controller without multi-agent coordination keeps tool permissions and step budgets directly inspectable; the control loop, tool registry, verifier, and evaluation harness are written in Python without an external agent framework. LuatCheck uses the 4B controller by default and supports a lighter 2B option; both are ternary-quantised Bonsai models[1] served by `llama.cpp`. All solid components in Figure 1 execute in the active pipeline; the orange modules (cross-encoder reranker and L1 citation verifier) are ablated in Section 5, whereas the dashed modules (dense vector index, semantic NLI check, answer memory, and trusted UI outbox) are implemented in code and switched off in the current ablation study.

**Execution trace and web interface.** Figure 2(a) traces a recorded trial on the labour-law benchmark (3 controller calls, 2 tool calls, 0 repairs), in which the controller searches, reads the top-ranked article, and produces a cited answer accepted by the verifier. Figure 2(b) shows the live web interface for the same query, displaying the answer, verification status, highlighted source passage, and step timeline.

[1] Bonsai 4B and 2B, ternary weights (about 2.1 bits per weight) in GGUF format, served locally by llama.cpp.

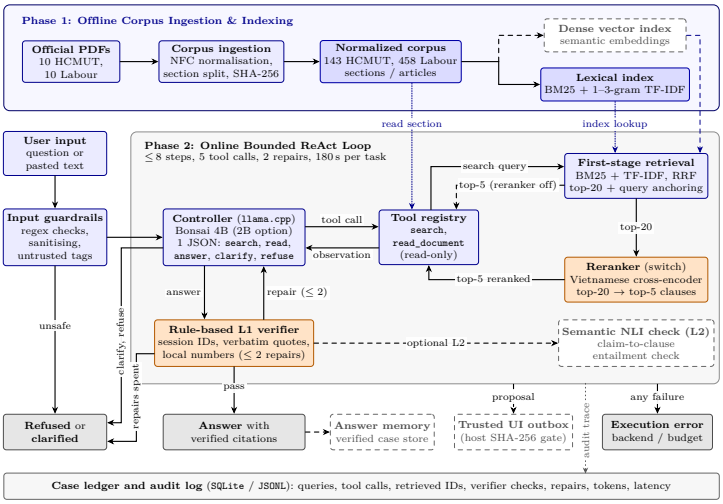



**Figure 1:** Two-phase architecture of LuatCheck: Phase 1 (top) ingests official PDFs into a normalized corpus and builds an in-memory BM25 + TF-IDF index; Phase 2 (main frame) runs the online bounded ReAct loop. Solid blue: core active modules. Solid orange: configurable modules ablated in Section 5. Dashed boxes/arrows: implemented in code, disabled or bypassed in D2 runs. Dotted blue arrows: read-only corpus/index lookups. Solid grey: terminal run states.

**Question** (translated): under Decree 293/2025, what is the hourly minimum wage in region IV?

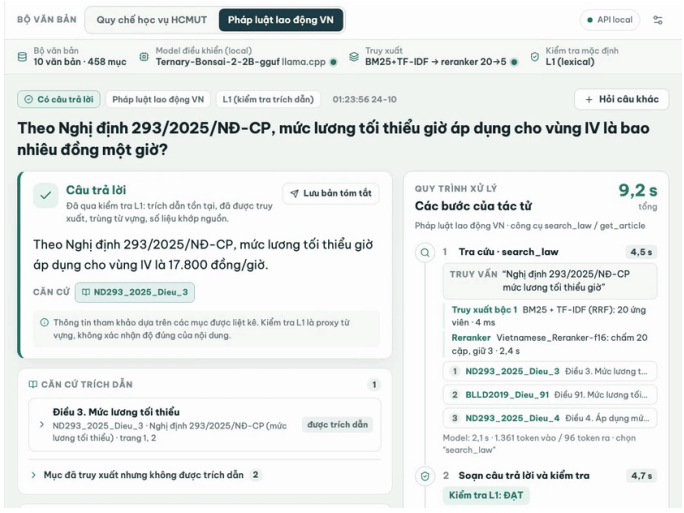
**1 search.** Query "Nghị định 293/2025/NĐ-CP mức lương tối thiểu giờ vùng IV"; BM25 + TF-IDF give 20 candidates, the reranker keeps 5; Article 3 of the decree is ranked first. *1,574 tokens in, 105 out.*

**2 read.** Full text of Article 3 of Decree 293/2025. *2,921 in, 166 out.*

**3 answer.** "17,800 VND per hour", quote "Vùng IV 3.700.000 17.800" from Article 3. **Citation verifier: accepted, 0 repairs.** *3,623 in, 211 out.*

**Outcome:** answered with a verified citation; 8,118 tokens in, 482 out in total. The answer equals the benchmark's reference answer.

(a) Recorded trial trace (4B, full system).




(b) Web interface, same question.

**Figure 2:** (a) One completed trial from the evaluation (a labour-law question about the hourly minimum wage, seed 42). (b) The web interface answering the same question, showing the answer, citation verifier status, highlighted source section and step timeline.

## 3 Tools and Permissions

Table 2 lists every tool and model-facing interface with its implementation status and permission level. The controller's action schema contains no write action, leaving no direct write path from model output. All non-read-only interfaces are restricted or gated in Python rather than by prompt instructions alone. Although the main agent loop calls the Python tool registry directly, an accompanying Model Context Protocol (MCP) server exposes the same interfaces and is tested for tool discovery, retrieval, blocked unverified writes, and proposal-only summaries. The demo serves a single local user per process; multi-user isolation is not certified.

| Interface | State | Permission level | Enforced boundary |
|---|---|---|---|
| Corpus search and read tools | Implemented, used | Read-only | Return local corpus text only. |
| Reranker (inside search) | Implemented, used | Read-only model component | Scores candidates; a failure is an execution error, never a silent fallback. |
| Answer-memory write | Implemented, off in all runs | Reversible write | Local database write only after the verification gate. |
| Case-saving tool of the MCP server | Implemented, blocked | Reversible write | A model caller cannot self-attest verification; the call returns blocked. |
| Summary-sending tool | Proposal only | Irreversible write | The model-facing tool only creates a proposal; it cannot approve or execute. |
| Trusted-UI outbox | Implemented | Irreversible write (local queue) | Needs host approval and a token bound to the payload hash; a queue receipt is not a delivery. |
| External sender, code execution | Not implemented | Irreversible write, code execution | No e-mail or webhook transport and no code-running tool exists. |

**Table 2:** Tool inventory with permission levels as classified in code.

## 4 Memory and Knowledge Sources

**Short-term context.** Each run maintains a single message list: the domain system prompt, the user question (and any pasted document) wrapped as untrusted input, followed by alternating action and observation turns. Tool observations are compacted: a search returns the best-matching clauses of each record with a truncation flag, a repeated query or document read returns a brief notice, and step-budget notices prompt the controller to conclude with an answer or refusal before reaching the step limit. The controller uses an 8,192-token context window, and a case ledger records queries, retrieved IDs, repairs, token counts, and latencies.

**Long-term memory (implemented, disabled in all benchmark runs).** Verified answers can be stored in a local SQLite database after passing the verification gate. Reuse requires a matching index version and a question similarity of at least 0.85, and any retrieved answer is re-verified before return. Long-term memory is disabled across all benchmark runs and in the live demo so that trials remain independent and cached answers cannot ignore case-specific facts. Before any database write, regex filters screen out personal identifiers.

**Knowledge sources.** Both corpora were built from public PDFs by text extraction, Unicode NFC normalisation, and segmentation at article or section headings. Each record stores its source URL, SHA-256 file hash, and page numbers, with each article or section serving as the retrieval unit. The HCMUT corpus comprises 10 documents and 143 records (five documents are amendment-only, image-only base regulations are omitted, and decision numbers come from a web listing). The labour-law corpus comprises 10 documents and 458 records (Decree 145/2020 and unnumbered annexes are omitted, and texts reflect their promulgated versions without tracking later amendments). Neither corpus has been reviewed by an external legal or academic-affairs specialist.

**Retrieval pipeline (current D2 system).** At start-up, LuatCheck builds in-memory BM25 (over title and content) and word 1–3-gram TF-IDF indices and combines their rankings with Reciprocal Rank Fusion (RRF). The top 20 candidates are rescored by a local Vietnamese cross-encoder reranker to keep the top 5 sections (articles longer than 1,500 characters are scored by their best overlapping window). To prevent poor model query rewrites from hurting recall, the first search of every run combines the user's original question with the model's query.

**Preliminary vector-search experiment (toward D3).** For the D3 retrieval study, we also integrated an optional persistent index that pairs SQLite FTS5 keyword search with `EmbeddingGemma-2` dense embeddings (270M text parameters, `bfloat16` on Apple Metal, ≈517 MB in RAM, $< 6$ MB on disk) into the codebase and demo UI. Testing this module on the two D2 corpora and an 818-query Vietnamese legal test set from Zalo AI (447 gold articles plus 553 BM25 hard negatives) showed three practical trade-offs:

- **Pure embedding search lags behind keyword search on legal text:** 768-d vector search alone reaches 52.9% Recall@1 (66.1% MRR@10) on the Zalo legal set, compared with 62.5% (75.4% MRR@10) for BM25+TF-IDF and 68.8% (80.2% MRR@10) for SQLite FTS5. Legal questions often depend on exact decree numbers, article IDs, or numeric thresholds, whereas dense embeddings pull in nearby clauses that share the same general topic.
- **Truncating embeddings from 768-d to 128-d hurts accuracy without saving time:** Matryoshka 128-d truncation drops vector Recall@1 by 13.0 percentage points (to 40.0%) and Recall@5 by 14.7 points (to 69.0%), while query latency barely changes because the transformer still runs the same forward pass.
- **Weighting keyword search higher in hybrid RRF gives the best recall:** An equal 50/50 RRF mix slightly lowers Recall@1 (to 67.0%) because approximate vector matches can outrank an exact keyword hit. Weighting FTS5 at 0.75 and 768-d vectors at 0.25 keeps exact keyword matches on top while still helping on paraphrased questions, reaching 69.1% Recall@1, 97.0% Recall@5, and 80.9% MRR@10 on the Zalo set (and 94.1–95.8% MRR@10 on the D2 corpora).

## 5 Evaluation Set and First Results

**Evaluation objective and D2 scope.** The primary objective of the evaluation in this D2 report is to demonstrate that the end-to-end agentic pipeline and automated evaluation harness built by the team run reliably and produce reproducible, auditable baseline measurements across configurations. Further experiments planned for **D3** (Section 7) will evaluate harder unseen multi-hop queries where lexical retrieval is not saturated, reduce verifier over-refusals under the 2-repair budget, benchmark dense retrieval, and test adversarial prompt injections.

**Benchmarks.** Each self-built benchmark has 50 Vietnamese questions over its corpus (100 now; D3 adds 50–100 unseen harder questions and ≥30 attack cases, Section 7): 40 single-record, 5 two-record and 5 out-of-scope questions to refuse. Each item includes gold record IDs, a reference answer, and verbatim quotes, locked by SHA-256 hash prior to data collection.

Candidate questions were drafted with an LLM, cross-checked by two independent LLM verifiers and an adjudicator, and validated programmatically for exact quote matching against the corpus, before Võ Phạm Tuấn Dũng manually reviewed all 100 items against the source PDFs on 23 October. Because the questions were synthesized from regulatory provisions rather than collected from live users, our post-hoc quality audit noted three limitations despite zero gold-label errors:

- plain BM25 on the raw question already ranks a gold record first for 89% (HCMUT regulations) and 78% (labour law) of questions, so retrieval cannot separate the configurations;
- about one third of the answerable questions can be answered without the corpus (LLM estimate);
- 22 HCMUT questions pin a document by a decision number taken from a web listing.

**Evaluation metrics.** *Strict success* counts a trial only if the run completed, the action was appropriate (answer when the corpus answers, refuse when it does not), the answer is judged correct and complete, and every factual claim is supported by the cited sources. Execution errors and failed judgments count

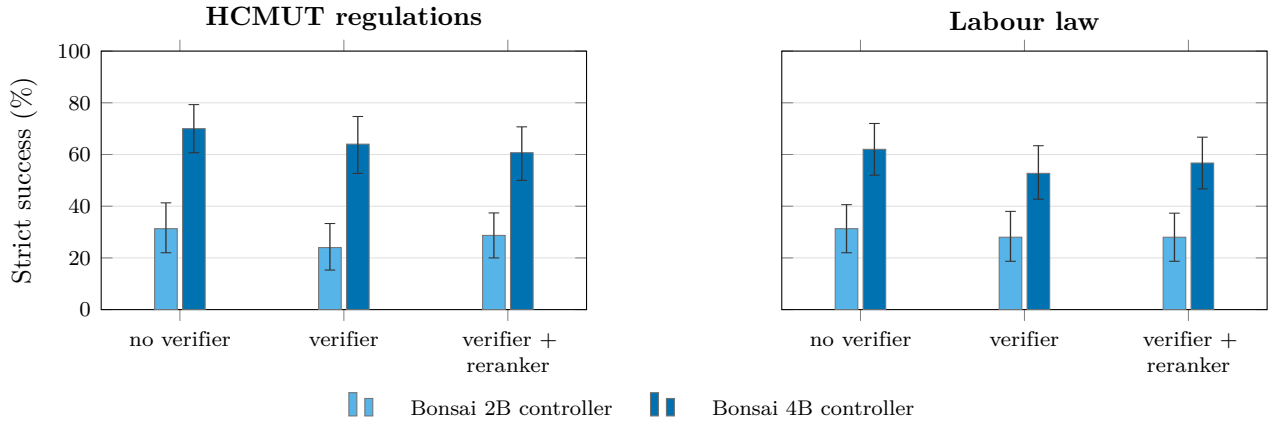



**Figure 3:** Strict success (LLM-judge lower bound; bars: mean over three seeds; whiskers: 95% bootstrap interval over the 50 questions) across the three configurations for the Bonsai 2B and Bonsai 4B controllers. The right-most group, *verifier + reranker*, is the full system.

as failures, so the reported rate is a conservative lower bound over all planned trials. Strict success is scored by the offline LLM judge and calibrated against our blind human audit below.

Three behavioural metrics require no LLM judge: behaviour match, over-refusal rate, and unsafe answer rate. Retrieval Recall@5 (whether gold articles appear in retrieved results) and citation F1 (overlap between cited and gold article IDs) evaluate document-level grounding independently of answer correctness. All confidence intervals are 95% bootstrap intervals over the 50 questions after averaging across the three seeds; at $N = 50$, differences under roughly 10 percentage points fall within sampling noise.

**LLM judge and human calibration audit.** An LLM judge (DeepSeek V4 Flash, accessed via an API gateway) scores each output blind to the system configuration and verifier outcome (offline only; model revision not pinned [2]; LLM judges can exhibit systematic biases [9]). On a blind 60-item human audit (30 per benchmark, 4B outputs), strict-success agreement is 96.7% (29/30, $\kappa = 0.93$; weighted 98.8%, $\kappa_w = 0.97$) on HCMUT and 90.0% (27/30, $\kappa = 0.80$; weighted 97.5%, $\kappa_w = 0.95$) on labour law; behaviour agreement is 100.0% and 93.3% ($\kappa = 0.84$). All four disagreements are documented in the repository. Separating correctness from citation support follows Gao et al. [3].

**Experimental setup and baselines.** The full system (verifier and reranker) is compared against two ablations: *no verifier* (neither verifier nor reranker) and *verifier* (verifier with repair, no reranker). Each configuration runs across three random seeds on all 50 questions per benchmark.

Two single-seed diagnostic baselines bracket the system configurations: *closed-book* (question only, no tools or corpus access) and *oracle* (gold records provided directly in context, evaluated on the 45 answerable questions). Each controller size is therefore evaluated across 900 system trials and 190 baseline trials (1,090 logged trials per controller). Both the 4B controller (default) and the 2B controller (lightweight option) are evaluated under the same protocol. Accuracy runs were collected on a shared server GPU (NVIDIA H100), while interactive latency was measured sequentially on an Apple M4 laptop (16 GiB).

**(1) Controller scale drives answer synthesis and citation quality (Figure 3).** For the full system (*verifier + reranker*), strict success is 60.7% [50–71] on HCMUT regulations and 56.7% [47–67] on labour law with the 4B controller, compared with 28.7% [20–37] and 28.0% [19–37] with the 2B controller.

Scaling the controller from 2B to 4B on the same pipeline adds +32.0 percentage points [21.3, 42.0] on HCMUT and +28.7 [18.0, 39.3] on labour law for the full system (+38.7 and +30.7 without verifier; +40.0 and +24.7 with verifier only). All six paired bootstrap differences across the three configurations and two benchmarks exclude zero.

**(2) Effect of the rule-based citation verifier on citation integrity and audited accuracy.** In Table 3, under the LLM judge alone (*judge-only strict success*, which grades answer prose without verifying exact substrings against the corpus), *No verifier* reaches 70.0% and 62.0% compared with

60.7% and 56.7% for the full system, because the verifier converts unresolved citation errors after two repair attempts into refusals.

Table 4 shows why: without the verifier, 25.6% (31/121) of HCMUT answers and 28.0% (37/132) of labour-law answers from the 4B controller contain unverifiable citations—such as altered quotes, unretrieved section IDs, or numbers copied from adjacent clauses (39.8% and 45.9% on the 2B controller). Because the LLM judge does not verify exact substrings or local-window numbers against the full PDF corpus, it still marks 22 (HCMUT) and 16 (labour law) of those answers as strictly successful. When we require both LLM-judge correctness and a passing L1 citation check (*Audited strict* in Table 4), enabling the verifier reduces unverified citations to 0.0%, repairs 17–20 rejected drafts per benchmark, and raises audited strict accuracy from 55.3% to 64.0% (+8.7 pp, Verifier) and 60.7% (+5.4 pp, Full) on HCMUT, and from 51.3% to 56.7% (+5.4 pp, Full) on labour law. Because first-stage Recall@5 is already 0.98–0.99 on these 100 questions, we keep both components and will test the reranker on a harder unseen benchmark in Week 45.

| | HCMUT regulations | | Labour law | |
|---|---|---|---|---|
| Configuration (4B controller) | All 50 questions | 45 answerable | All 50 questions | 45 answerable |
| Closed-book (no documents) | 2.0 [0–6] | 0.0 [0–0] | 24.0 [12–36] | 24.4 [13–38] |
| No verifier, no reranker | 70.0 [61–79] | 69.6 [60–79] | 62.0 [52–72] | 60.7 [50–71] |
| Verifier (with repair) | 64.0 [53–75] | 61.5 [50–73] | 52.7 [43–63] | 51.1 [40–62] |
| **Verifier + reranker (full system)** | **60.7 [50–71]** | 59.3 [49–70] | **56.7 [47–67]** | 54.1 [43–64] |
| Oracle (gold records given up front) | – | 82.2 [71–93] | – | 75.6 [62–87] |

**Table 3:** Strict success (%, LLM-judge lower bound, 95% bootstrap interval over questions) of the D2 system with the 4B controller. Configurations: three seeds; closed-book and oracle: one seed. The oracle runs only on the 45 answerable questions, so the right-hand columns of each benchmark compare every row on the same 45 answerable questions.

| Benchmark | Configuration | Unverified cit. leak (%) | Audited strict (%) | Behaviour match (%) | Over-refusal (%) | Faithfulness (%) | Citation F1 | Recall@5 |
|---|---|---|---|---|---|---|---|---|
| HCMUT regulations | No verifier | 25.6 | 55.3 | 88.0 | 9.8 | 89.8 | 0.820 | 0.989 |
| | Verifier | **0.0** | **64.0** | 82.7 | 17.4 | 90.3 | 0.704 | 0.989 |
| | Verifier + reranker | **0.0** | **60.7** | 80.0 | 18.9 | 89.3 | 0.714 | 0.978 |
| Labour law | No verifier | 28.0 | 51.3 | 94.0 | 4.4 | 87.3 | 0.841 | 0.989 |
| | Verifier | **0.0** | 52.7 | 81.3 | 16.8 | 86.3 | 0.728 | 0.989 |
| | Verifier + reranker | **0.0** | **56.7** | 79.3 | 19.1 | 85.6 | 0.691 | 0.993 |

**Table 4:** Ablation of the rule-based citation verifier and the reranker (4B controller, 3 seeds × 50 questions per row). *Unverified cit. leak*: share of emitted answers whose quotes, section IDs or numbers fail deterministic source verification against the full PDF corpus (31/121 HCMUT and 37/132 labour answers in *No verifier* vs 0/109 on HCMUT and 0/111–108 on labour law with the verifier). *Audited strict*: trials that pass both the LLM judge and deterministic L1 citation verification. *Behaviour match* (deterministic action-match rate), *Over-refusal*, *Citation F1* and *Recall@5* require no LLM judge.

**(3) Baselines.** Without corpus access, the 4B model scores 2.0% [0–6] on HCMUT regulations and 24.0% [12–36] on labour law, indicating that pre-training knowledge contributes partially on general labour law. When provided with the gold records directly, the oracle baseline reaches 82.2% [71–93] and 75.6% [62–87] on the 45 answerable questions (compared with 69.6% and 60.7% for *No verifier*), showing that roughly 18–24% of answerable questions still fail at multi-clause synthesis even under ideal retrieval. Across all 1,090 trials, 17 ended as execution errors (wall-time limit) and are counted as failures.

**Error analysis and experimental roadmap for D3.** Three failure modes account for the remaining errors in the 4B full system: (i) *repair-budget refusals* (13/150 HCMUT and 18/150 labour trials refused after two L1 quote/number rejections on answerable questions), (ii) *multi-clause synthesis omissions* (oracle gap of 18–24% when gold articles are provided), and (iii) *wall-time budget timeouts* (17 of 1,090 trials). Because D2 evaluation focuses on establishing a working, measurable baseline and lexical

retrieval is near ceiling on this initial 100-question set, D3 will target these bottlenecks on harder unseen multi-hop and paraphrased benchmarks, refine the repair loop, and execute entry-point attack tests.

**Cost, memory footprint, and latency.** The controller and reranker run locally at zero API cost; offline LLM judging across the two 545-trial runs cost USD 1.06 and USD 1.10. On the 16 GiB Apple M4 laptop, the active baseline stack uses about 4.8 GB of resident RAM (RSS) with the 4B controller (3.4 GB for `llama-server`, 1.2 GB for the reranker, and 0.2 GB for the FastAPI app and lexical indices) and 2.8 GB with the 2B controller (1.4 GB for the 2B server). Enabling dense vector search with `EmbeddingGemma-2` (`bfloat16`) adds about 0.5 GB (517 MB of model weights), bringing total RAM to roughly 5.3 GB on 4B and 3.3 GB on 2B. Per trial, the 4B full system reads an average of 12,070 (HCMUT) and 11,477 (labour law) input tokens and generates 763 and 865 output tokens across 3.4 and 3.7 model calls. Full accuracy runs took 34 and 33 minutes of wall time on the shared H100. In sequential single-query runs on the Apple M4 laptop (50 questions per benchmark), median response time for the full system is 33.0 s (p95 86 s) on HCMUT and 29.8 s (p95 72 s) on labour law with the 4B controller ($\approx$ 43 tokens/s), and 13.3 s (p95 30 s) and 14.6 s (p95 28 s) with the 2B controller ($\approx$ 92 tokens/s).

**Reproducibility.** Repository: https://github.com/tuandung222/LuatCheck-D2; reported commit `0928bc5fc437`. All numbers and tables are regenerated from the recorded runs by one command (`make report-check`).

## 6 Threat Model

**Assets.** (A1) Faithfulness: answers and citations that users rely on must trace to source sections [7]. (A2) User data: questions, pasted documents, memory and the audit log. (A3) Integrity of corpora, domain profiles, prompts, model weights and the frozen benchmark. (A4) The tool boundary: no unreviewed write or egress. (A5) Availability of a bounded local machine.

**Attacker.** The attacker can submit arbitrary questions and pasted documents through the interface or API (instruction override, delimiter spoofing, prompt or memory exfiltration, requests to evade a regulation), including text copied from a hostile third party. The attacker cannot change code, corpora, configuration, weights, the ledger or the approval state of the trusted UI, and cannot reach loopback servers or other users' sessions. Non-goals: a compromised host or supply chain, poisoned official sources, multi-tenant isolation, and the correctness of the regulations themselves.

## 7 Weekly Plan to the First Final Presentation

Table 6 runs to the first final presentation in week 49. The final phase focuses on harder unseen multi-hop questions, real-user feedback from the HR walkthrough, and entry-point prompt-injection testing.

**Riskiest assumption and fallback.** The riskiest assumption is that a small 2–4B local controller maintains high strict accuracy on harder, unanchored questions (paraphrased scenarios that do not name the decision or decree number) without exceeding the 2-repair budget. If strict success drops sharply on the new benchmark or dense retrieval and NLI verification do not improve over the current pipeline, we keep the bounded loop, the L1 citation verifier, the two corpora and the evaluation harness as the core deliverable, report the measured numbers transparently, and drop the optional NLI check and extra domains.

**Self-assessment: on track for D3.** At D2, the end-to-end system, both corpora, the 100-question evaluation suite, the 60-item judge audit, and the one-command reproduction pipeline are complete and verified to run and measure reliably (57–61% strict success on 4B). Further improvements and broader experiments—harder unseen multi-hop benchmarks (W45–W46), verifier false-reject reduction, and red-team attack tests (W47)—are scheduled for **D3**.

| Entry point (asset) | Mitigation | Residual risk |
|---|---|---|
| Question, pasted document (A1, A4) | Regex guardrails (shared and per-domain patterns) before any model call, Unicode and delimiter sanitising, untrusted-text tags, fixed action space without a write action | Paraphrases bypass patterns; Vietnamese and English only; false positives; no learned classifier; not attack-tested yet. |
| Retrieved text (A1) | Observations are tagged as tool results; the citation verifier checks ids, provenance and quotes | Tagged, not filtered: an instruction inside a source document would reach the model. |
| Ingested documents (A3) | Offline, manual ingestion; source URL, hash and pages per record; benchmark locks hash the corpora | No signature check; trust in the official site. |
| Memory (A1, A2) | Off in all runs and in the demo; index version, similarity, re-verification, personal-data patterns | Poisoning by an earlier user; incomplete personal-data filter. |
| Weights, configuration, binaries (A3) | Recorded hashes and commit, identity checks before each run, loopback-only endpoints | Third-party weights and the llama.cpp fork are not audited. |
| Judge gateway (A2) | Offline only; benchmark data only; key kept in the environment; blind packets; output validation; human audit | Candidate text could steer the judge (untested); model version not fixed. |
| Web interface, API, outbox, long inputs (A4, A5) | Loopback bind, session cookie and CSRF token, size limits; proposal-only tools with host approval; step, tool, repair, time limits | Single-user assumption; limits turn overload into errors, not into defences. |
| Web pages | Not applicable: there is no browsing tool. | – |

**Table 5:** Threat model by entry point. Mitigations are implementation boundaries, not measured attack effectiveness. The full-system injection tests [1, 4] belong to the final report (planned for week 47; no attack numbers in this report).

| Week (dates) | Owner | Deliverable and acceptance |
|---|---|---|
| 24–25 Oct (Done) | Dũng, all | Blind human audit of the 60-item judge sample (30 per benchmark); agreement (96.7% and 90.0%) and $\kappa$ (0.93 and 0.80) added to Section 5. |
| Sun 25 Oct, Wed 28 Oct | Quyền (leader), all | Submit report and slides; presentation with live demo and backup video. |
| Week 45 (2–8 Nov) | Quyền, Phong | HR walkthrough with findings. Design and freeze a new benchmark of unseen, harder questions (scenario, multi-hop, conditional and temporal, paraphrased), reviewed by members. |
| Week 46 (9–15 Nov) | Dũng, Quyền | Run the current system and its ablations on the new benchmark; add the dense-retrieval ablation; failure analysis. |
| Week 47 (16–22 Nov) | Phong, Dũng | Entry-point attack tests; permission and fault checks. |
| Week 48 (23–29 Nov) | All | Reproduction from the recorded runs, demo hardening, final report. |
| Week 49 (30 Nov–2 Dec) | All | Tagged release and Round 1 presentation on 2 December. |

**Table 6:** Weekly plan to the first final presentation, with assigned owners.

## 8 Member Contributions and AI-Use Statement

| Member | Contrib. | Role in the proposal | Completed work and repository evidence |
|---|---|---|---|
| Lê Trí Quyền (Group Leader) | 33% | Retrieval and serving: ingestion, BM25 + dense index, RRF, reranking, local deployment | Corpus ingestion and normalisation, BM25 + TF-IDF and vector indexing with RRF fusion, cross-encoder reranker integration, `llama.cpp`/Prism local serving and cross-platform Docker deployment (commits `555919a`, `8fed7c5`, `15fd823`, `ad0881b`, `bdb5d69`, `415efda`). |
| Nguyễn Hoàng Phong | 33% | System and security: tool layer, UI, audit log, persistent store, fault and injection harness | Tool registry and code-enforced permissions, regex guardrails and PII filtering, Web and Streamlit interfaces with trusted outbox, SQLite store and audit ledger, and fault/security test suites (commits `6dc1dbc`, `78bb069`, `2c1f3d0`, `d117349`, `7ce7e45`, `1272dc7`, `e0b4a05`, `edc9bfa`). |
| Võ Phạm Tuấn Dũng | 33% | Agent, architecture and evaluation: system design, model selection, control loop, verifier and evaluation | Idea proposal and verification, system architecture and local model selection, codebase organisation and refactoring; bounded ReAct control loop and config-driven pipeline, L1 citation verifier and bounded repair, 100-task benchmarks, evaluation harness, GPU/Mac runs, 60-item judge audit, statistical analysis and D2 report/slides (all remaining commits in the repository). |

**Table 7:** Member contributions (equal 33% split) matched to proposal responsibilities and commit history.

**AI-use statement.** The group used a range of AI models from Anthropic (Claude), Google (Gemini, Gemma) and OpenAI (Codex) for discussion and execution support. The problem framing, system scope, design decisions, task assignment to coding agents, supervision and code review were done by the group; every change was reviewed and approved by a group member before it was committed. AI coding agents implemented code from specifications written or approved by the group. AI models were also used, under the group's direction, to draft and cross-check benchmark questions (all questions were then reviewed by a group member), to act as the evaluation judge (an LLM estimate, checked against the group's human audit), and to draft analyses and report text that the group edited.

**Verified by the group.** Võ Phạm Tuấn Dũng reviewed all 100 benchmark questions against the source PDFs (23 October), audited the 60-item blind judge sample (25 October: 96.7% agreement, $\kappa = 0.93$ on HCMUT; 90.0%, $\kappa = 0.80$ on labour law), verified the architecture and code across all commits, and checked the Mac demo runs. Lê Trí Quyền verified corpus extraction, BM25/TF-IDF indexing and local `llama.cpp`/reranker serving. Nguyễn Hoàng Phong verified tool permission boundaries, guardrails and Web UI states. Automated tests use real components and no mocks (425 passed, 2 skipped, 1 failed needing an unstarted Ollama backend).

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